Particle velocity, Phase velocity

& Group velocity

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Whatever we will discuss

- ✓ Definition:
  - a) Particle velocity
  - b) Phase velocity or wave velocity
  - c) Group velocity
- ✓ Relation between Particle velocity and Wave velocity
- ✓ Relation between Group velocity and Phase velocity
- ✓ Distinguish between Group velocity and Phase velocity
- ✓ Related mathematical problems

Particle Velocity

Equation of a simple harmonic plane progressive wave is

$$y = A \sin \frac{2\pi}{\lambda}(vt - x) \quad \ldots\ldots \quad \ldots\ldots.. \quad (1)$$

Differentiating the above equation with respect to time, we have particle velocity,

$$U = \frac{dy}{dt} = \frac{2\pi A v}{\lambda} \cos \frac{2\pi}{\lambda}(vt - x) \quad \ldots\ldots \quad \ldots\ldots.. \quad (2)$$

Maximum value of particle velocity $= \dfrac{2\pi A}{\lambda} \times (wave\ velocity)$

Particle Acceleration

The acceleration of the particle is

$$\frac{d^2y}{dt^2} = -\frac{4\pi^2Av^2}{\lambda^2}\ \sin\ \frac{2\pi}{\lambda}(vt-x)$$

$$= -\frac{4\pi^2v^2}{\lambda^2}\left[\ A\ \sin\ \frac{2\pi}{\lambda}(vt-x)\right]$$

$$= -\left(\frac{4\pi^2v^2}{\lambda^2}\right).y$$

The maximum acceleration (at y=A) $= -\dfrac{4\pi^2v^2}{\lambda^2}.A$

Ref.: Physics for Engineers (Part-1) -Dr. Gias Uddin Ahmed; pp: 155-156

Particle velocity and wave velocity

$$U = \frac{dy}{dt} = \frac{2\pi A v}{\lambda} \cos \frac{2\pi}{\lambda}(vt - x) \quad \cdots\cdots \quad \cdots\cdots\cdots \quad (2)$$

Differentiating eqn. (1) with respect to x, to get the slope of displacement curve

$$\frac{dy}{dx} = -\frac{2\pi A}{\lambda} \cos \frac{2\pi}{\lambda}(vt - x) \quad \cdots\cdots \quad \cdots\cdots\cdots \quad (3)$$

Comparing eqns. (2) and (3), we get

$$U = \frac{dy}{dt} = -v.\frac{dy}{dx}$$

Thus, particle velocity at a point = - (wave velocity) x (slope of the displacement curve at that point

Phase velocity or wave velocity

Waves can be in a group and such groups are called wave packets, so the velocity with which a wave packet travels is called group velocity. **The velocity with which the phase of a wave travels is called phase velocity**.

The wave velocity is also called the phase velocity v_p

$$v_p = \frac{\omega}{k}$$

The velocity of propagation of a wave is the velocity with which a displacement of a given phase moves forward.

If a wave has angular frequency ω and wave number k, then a point moves with the velocity ω/k called the phase velocity and is represented by v_p

$$Phase\,velocity,\ v_p = \frac{\omega}{k}$$ where, $\omega = 2\pi\upsilon$ and $k = 2\pi/\lambda$

Group velocity

When large number of plane waves of slightly different wave lengths and frequencies travel in the same direction, groups of waves are formed. Those groups of waves are known as wave packets. Each group of waves then moves with a velocity known as group velocity. The group velocity is defined as in mathematically, $v_g = \dfrac{d\omega}{dk}$.

If phase velocity of wave depends upon the wave length, then group velocity is different than phase velocity and shape of the group of waves changes as it moves forward.

Relation between the Group velocity and the Phase velocity

We know the wave number or propagation constant is

$$k = \frac{2\pi}{\lambda}; \;\; \therefore \lambda = \frac{2\pi}{k}$$

and the phase velocity, $v_p = \dfrac{\omega}{k}; \;\; \therefore \omega = v_p k$

and the group velocity, $v_g = \dfrac{d\omega}{dk} = \dfrac{d}{dk}(v_p k) = v_p + k\dfrac{d\,v_p}{dk}$

$$= v_p + k\frac{d\,v_p}{d\lambda} \cdot \frac{d\lambda}{dk}$$

Relation between the Group velocity and the Phase velocity

$$= v_p + k \frac{d\, v_p}{d\lambda} \cdot \frac{d\lambda}{dk}$$

$$But \;\; \frac{d\lambda}{dk} = \frac{d}{dk}\left(\frac{2\pi}{k}\right) = -\frac{2\pi}{k^2}$$

$$v_g = v_p - \frac{2\pi}{k}\frac{d\, v_p}{d\lambda} = v_p - \lambda \frac{d\, v_p}{d\lambda}$$

This equation shows that v_g is less than v_p when the medium is dispersive, i.e., when v_p is a function of λ.

Relation between the Group velocity and the Phase velocity

$$v_g = v_p - \lambda \frac{d\,v_p}{d\lambda}$$

If the medium is not dispersive, i.e., when the waves of all wavelength travels with the same speed in the medium then

$$\frac{d\,v_p}{d\lambda} = 0$$

$$\textit{Then the relation becomes, } v_g = v_p$$

Therefore, when the medium is not dispersive then the group velocity is also called phase velocity.

Distinguish between Group velocity and Phase velocity

S/N	Phase Velocity	Group Velocity
1.	The Phase velocity refers to the velocity of a monochromatic wave.	Group velocity is the velocity with which the entire group of waves would travel.
2.	Phase velocity, $v_p = \omega/k$	Group Velocity, $v_g = d\omega/dk$
3.	It transmits the phase.	It transmits the energy.
4.	Signal are not transmitted with the phase velocity.	Signal are transmitted with the group velocity.

Thank You

Thanks for your attention